

The Impact of Neurocysticercosis in California: A Review of Hospitalized Cases



To assess the burden of neurocysticercosis (NCC) in California we examined statewide hospital discharge data for 2009. There were 304 cases hospitalized with NCC identified (incidence = 0.8 per 100,000). Cases were mostly Latino (84.9%), slightly more likely to be male than female (men 57.6%, women 42.4%) with an average age of 43.5 years. A majority of cases were hospitalized in Southern California (72.1%) and many were hospitalized in Los Angeles County (44.7%). Men were more likely than women to have severe disease including hydrocephalus (29.7% vs. 18.6%, $p = 0.027$), resulting in longer hospitalizations (>4 days, 48.0% vs. 32.6%, $p = 0.007$) that were more costly (charge $> \$40$ thousand men = 46.9% vs. woman = 4.1%, $p = 0.026$). Six deaths were recorded (2.0%). The total of NCC-related hospital charges exceeded \$17 million; estimated hospital costs exceeded \$5 million. Neurocysticercosis causes appreciable disease and exacts a considerable economic burden in California. Author Summary Neurocysticercosis (NCC) is considered one of the major neglected infections of poverty in the United States, with mortality studies indicating that California bears the highest burden of this disease. Although NCC is a reportable disease in California, studies indicate that this disease goes largely under-reported, contributing to the lack of information about the disease distribution and burden. In this manuscript, we reviewed the distribution of NCC hospitalizations in California, demographics of those hospitalized and total hospital-related charges for 2009. This study revealed that a majority of persons hospitalized with NCC in California receive their medical service in Southern California hospitals, primarily in the County of Los Angeles. As compared to women hospitalized for this disease, men had a longer and more costly hospitalization with more severe symptoms such as hydrocephalus, a diagnosis suggestive of extraparenchymal infection. The reasons for this difference in NCC severity by gender are not clear, but do not appear to be due to delay in seeking medical care or a language barrier. The intensity of hospital care needed to manage these cases and the sizable NCC hospitalization charge underscores the considerable economic burden this disease presents in California.